- The probit model is another nonlinear regression model where the dependent variable is a binary variable.
- The coefficients in the probit model can be interpreted as the change in the probability of the dependent variable for a one-unit change in the independent variable.
- The probit model is like the logit model, but it assumes that the error term follows a normal distribution instead of a logistic distribution.

## 15.8 KEY WORDS

**Binary Model** : A binary model is a statistical model that predicts a categorical dependent variable with two possible outcomes. The dependent variable is typically coded as a binary variable, such as 0 or 1, or 'yes' or 'no'. The independent variables can be continuous or categorical.

**Qualitative Dependent Variable** : A qualitative dependent variable is a variable that can take on one of a limited number of categories, such as yes or no, high or low, or agree or disagree. It is a type of categorical variable. The qualitative variable is not continuous.

**Dichotomous Variables** : A dichotomous variable is a variable that can only take on two values, such as "yes" or "no, " "true" or "false, " or "male" or "female." Dichotomous variables are often used in surveys and experiments, where the goal is to understand the relationship between two or more variables.

**Odds Ratio** : It is calculated by dividing the odds of occurrence of an event by the odds of the event occurring in another group. Odds ratios are always positive. Odds ratios can be greater than 1 or less than 1. An odds ratio of 1 indicates no association between the two variables. An odds ratio greater than 1 indicates that the event is more likely to occur in the first group than in the second group. An odds ratio less than 1 indicates that the event is less likely to occur in the first group than in the second group.

**Marginal Effect** : Marginal effects are calculated at a *specific point* in the data. The point at which the marginal effect is calculated is called the base point. Marginal effects can be *positive or negative*. A positive marginal effect indicates that an increase in the independent variable is associated with an increase in the dependent variable. A negative marginal effect indicates that an increase in the independent variable is associated with a decrease in the dependent variable. Marginal effects can be interpreted in different ways depending on the *context*. For example, a marginal effect of

0.10 in a logistic regression model may be interpreted as a 10% increase in the probability of the event occurring.



## 15.9 SOME USEFUL BOOKS

Baltagi. Badi H. (2008), *Econometrics*, Fourth Edition, Springer-Verlag Berlin Heidelberg.

Berndt, Ernst R. (1991), *The Practice of Econometrics: Classic and Contemporary*, Addison-Wesley.

Cameron, Colin and Pravin Trivedi (2005), *Microeconometrics*, Cambridge: Cambridge University.

Maddala, G. S. (1983), *Limited Dependent and Qualitative Variables in Econometrics*, Cambridge University Press, Cambridge.

Verbeek, Marno (2017), 'A *Guide to Modern Econometrics*,' Fifth Edition, John Wiley and Sons Ltd.

Wooldrige, Jeffrey (2022), *Introductory Econometrics*, Seventh Edition, Cengage Learning

## 15.10 ANSWERS / HINTS TO CHECK YOURPROGRESS EXERCISES

**Check Your Progress 1**

1) (a) Since all t-values (coefficient divided by standard error) are greater than 2, all the coefficients are statistically significant at 5 per cent level of significance.

(b) The intercept term gives the "probability" that a female person (since $D_i$= 0 for female) with zero income will own a house. This value is negative. Since probability cannot be negative, we may treat this value as zero.

(c) The slope value of 0.0017 means that for a unit change in income, on average, the probability of owning a house increases by 0.0017 or about 0. 17 percent. Of course, given a particular level of income, we can estimate the actual probability of owning a house. If, for example, income = Rs. 500, the estimated probability of owning a house is

$$(\hat{Y}_i|X_i = 500, D_i = 1) = -0.084 + 0.0017 * 500 + 0.008 * 1$$

$$= 0.846 = 84.6 \text{ percent.}$$

$$(\hat{Y}_i|X_i = 500, D_i = 0) = -0.084 + 0.0017 * 500 + 0.008 * 0$$

$$= 0.766 = 76.6 \text{ percent.}$$

(d) The coefficient of variable $D_i$ is 0.008. It implies that for a male (when $D_i = 1$), the intercept increases by 0.008. Therefore, the probability of owning a house for males is greater than that for females by 0.8 percent.

2) Go through Section 15.2.2. You have to summarise the points given there and answer.

**Check Your Progress 2**

1) The log of odds-ratio is given by $L_i = \ln\left(\frac{p_i}{1-p_i}\right) = \alpha_0 + \alpha_1 X_i + u_i$.

2) You can interpret the results of the logit model through its Logit, Odds-ratio and marginal effect. See Sub-Section 15.3.2.

3) Go through Sub-Section 15.3.2 and answer.

**Check Your Progress 3**

1) Go through Section 15.4 and answer.

2) Go through Section 15.6 and answer.

# UNIT 16 INTRODUCTION TO SIMULTANEOUS EQUATIONS MODELS*

**Structure**



## 16.0 OBJECTIVES

After going this Unit, you will be able to

- analyse the nature of simultaneous relationship between variables;
- distinguish between endogenous variables and pre-determined variables;
- explain the concept of simultaneity bias and its implications;
- examine the identification status of an equation in a simultaneous equations system; and
- applythe order and rank conditions for identification of an equation.

*Adapted from Units 13 and 14 of IGNOU SLM, MECE 001: Econometric Methods, written by Prof. Biswajit Nag and Ms. Satavisha Mukherjee, Indian Institute of Foreign Trade (IIFT), New Delhi

## 16.1 INTRODUCTION

The econometric models we studied so far talk about single equation regression models where a dependent variable is related to a set of independent or explanatory variables. In all these regression models, $y$ is the 'dependent' or 'explained' variable and $x_1$, $x_2$, … are the 'independent' or 'explanatory' variables. Thus $y$ depends upon $x$, but the reverse is not true. In real life, however, we normally find variables that are dependent on each other. In this sort of relationship, the equilibrium values of the variables can be determined by considering the simultaneity in relationship.

In simultaneous relationships a single equation under investigation is part of a wider phenomenon. In such cases the variables can be explained with the help of a system of equations. Let us take an example. At the macro level, as you know from macroeconomics course, aggregate consumption expenditure depends upon aggregate disposable income. Aggregate disposable income, on the other hand, depends upon the national income and taxes imposed by the government. Moreover, national income is dependent on aggregate consumption expenditure of the economy. Thus, estimation of a single equation between aggregate consumption and disposable income will not provide us with consistent and unbiased estimators.

## 16.2 SOME EXAMPLES OF SIMULTANEOUS EQUATIONS MODELS

In regression models one of the crucial assumptions is that the exogenous variables $x$'s are independent of the error term $u$ (see Unit 4). This assumption is violated in simultaneous relationships. Let us illustrate this through the demand and supply model.

We can write the demand function as

$$q = \alpha + \beta p + u \qquad \ldots(16.1)$$

where $q$ represents the quantity demanded, $p$ the price, and $u$ the stochastic error term, representing random shifts in the demand function.

From microeconomic theory we know that the equilibrium price and quantity are determined in the market by intersection of supply and demand curves (see Fig. 16.1). Therefore, we cannot determine equilibrium price by solving the demand equation independently. Again, if there is a random shift in the demand curve we see that it will affect both equilibrium price and quantity (considering the normal slope of supply curve as in panel-a of Fig. 16.1). In case the supply curve is vertical, then a shift in the demand curve affects equilibrium price only (see panel-b of Fig. 16.1). Again, if the supply curve is horizontal, a shift the demand curve affects equilibrium quantity only keeping the level of equilibrium price unchanged (see panel-c of Fig. 16.1). Thus we see that demand curve alone cannot determine the equilibrium price and quantity. Moreover, the slope of the supply curve influences the relationship between price and quantity. If we consider the demand equation only, the slope of the supply curve gets included in the term '$u$'. The inclusion of the slope of supply curve in the error term leads to correlation between price and error term, which violates one of the basic assumptions of OLS, that is, $cov(Xu) = 0$.So, if we ignore the supply curve and estimate the demand equation (16.1) by OLS, then it will produce inconsistent estimates of the parameters.

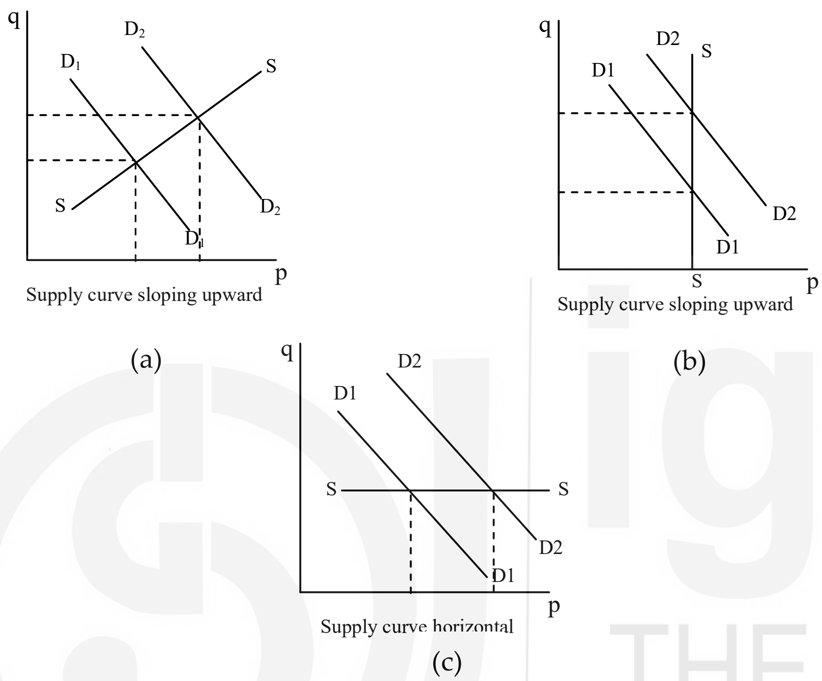


**Fig. 16.1: Interaction between Supply and Demand Curves**

The problem is that in the demand-supply model we cannot consider the demand function in isolation when we study the relationship between quantity and price as in equation (16.1).The solution is to bring the supply function into picture; and estimate the demand function and the supply function together. Such models are known as simultaneous equations models.

An econometric model is based on certain theoretical concepts. Thus, the simultaneous equations model (SEM) can be best understood by some practical examples. Here we present three examples of SEM.

**Example 16.1**

*Wage-Price Model*

The model of money wage and price determination is a good example of SEM. According to the model wage rate is determined by unemployment rate in the economy and the rate of price change (that is, inflation) in an unrestricted labour market (reflected in equation (16.2)). The rate of inflation, on the other hand, is a function of the change in wage rate and money supply (reflected in equation (16.3)). The model consists of the two equations as follows:

$$W_t = \alpha_0 + \alpha_1 U_t + \alpha_2 P_1 + u_{1t} \qquad \text{…(16.2)}$$

$$P_t = \beta_0 + \beta_1 W_t + \beta_2 R_1 + \beta_3 M_t + u_{2t} \qquad \text{…(16.3)}$$

where $W$ = rate of change in money wage

$U$ = unemployment rate in percentage

$P$ = rate of change in prices

$R$ = rate of change in cost of capital

$M$ = supply of money

$t$ = time

$u_1, u_2$ = stochastic disturbances

As the price variable $P$enters the wage equation (16.2) and the wage variable $W$ enters the price equation (16.3), both the two variables are inter-dependent. Moreover, the explanatory variables$P$ in (16.2) and $W$ in (16.3) are stochastic in nature. Recall that in OLS it is assumed that the explanatory variables are not stochastic. Therefore, these stochastic explanatory variables are expectedly correlated with the stochastic disturbance terms. This correlation violates the assumption of OLS that error term is independent of independent variable. Thus, the classical OLS method is not applicable for estimation of the parameters of the two equations individually.

**Example 16.2**

*Keynesian Model of Income Determination*

Let us consider the simple Keynesian model of income determination:

Consumption function$C_t = \beta_0 + \beta_1 Y_t + u_t \quad 0 < \beta_1 < 1$ … (16.4)

Income identity: $Y_t = C_t + I_t (= S_t)$ … (16.5)

where $C$ = consumption expenditure

$Y$= income

$I$ = Investment (assumed to be exogenous)

$S$ = Savings

$t$ = time

$u$ = stochastic disturbance term

$\beta_0$and$\beta_1$ are parameters.

The parameter $\beta_1$ represents the marginal propensity to consume (MPC). From macroeconomic theory we know that $\beta_1$ is expected to remain between 0 and 1.Equation (16.4) is the consumption function which shows that consumption consists of two parts: autonomous consumption expenditure ($\beta_0$) and income induced consumption ($\beta_1 Y$). Equation (16.5) is the national income identity, signifying that total income is equal to total consumption expenditure plus total investment expenditure.

From the consumption function (16.4) we find that $C$ and $Y$ are interdependent and that$Y_t$ is not expected to be independent of the disturbance term, $u_t$. When $u_t$ changes (because of a variety of factors included in the error term) the consumption function shifts, which, in turn affects $Y_t$. Therefore, once again the classical least-squares method is not applicable to (16.4). If applied, the estimators thus obtained will be inconsistent.

**Example 16.3**

*Macroeconomic Model*

The celebrated IS model, or goods market equilibriummodel of macroeconomics (see Unit 3 of MEC-102) in its non-stochastic form can be expressed as

Consumption function: $C_t = \beta_0 + \beta_1 Y_{dt}$ $\quad 0 < \beta_1 < 1$ …(16.6)

Tax function: $T_t = \gamma_0 + \gamma_1 Y_t$ $\quad 0 < \gamma_1 < 1$ …(16.7)

Investment function: $I_t = \theta_0 + \theta_1 r_t$ …(16.8)

Disposable income: $Y_{dt} = Y_t - T_t$ …(16.9)

Government expenditure: $G_t = \bar{G}$ …(16.10)

National income identity: $Y_t = C_t + I_t + G_t$ …(16.11)

where $Y$ = national income

$C$ = consumption spending

$I$ = plannedor desired net investment

$G$ = Given level of government expenditure

$T$ = taxes

$Y_d$ = disposable income

$r$ = interest rate

In order to solve the above equation system, we first substitute (16.7) into (16.9); and then the value of $Y_{dt}$into (16.6). Subsequently, we put the resulting equation for $C$, and equation (16.8) for $I$ and (16.10) for $G$ into (16.11). Consequently, we obtain

$$Y_t = [\beta_0 + \beta_1(Y_t - T_t)] + [\theta_0 + \theta_1 r_t] + \bar{G}$$

By re-arranging terms in the above equation, we obtain

$$Y_t = \pi_0 + \pi_1 r_t \qquad \text{…(16.12)}$$

where

$$\pi_0 = \frac{\beta_0 - \alpha_0\beta_1 + y_0 + \overline{G}}{1 - \beta_1(1 - \alpha_1)}$$

and

$$\pi_1 = \frac{1}{1-\beta_1(1-\alpha_1)}$$

Equation (16.12) is the equation of the IS curve, which provides the combinations of the interest rate and level of income such that the goods market is in equilibrium.

What would happen if we were to estimate, say, the consumption function (16.6) in isolation?Could we obtain unbiased and/or consistent estimates of $\beta_0$and$\beta_1$?Such a result is unlikely because consumption depends on disposable income, which depends on national income$Y$.But $Y$depends on $r$ and $G$ as well as the other parameters entering in$\pi_1$. Therefore, unless we consider all these influences, a simple regression of $C$ on $Y_d$ is bound to give biased and/or inconsistent estimates of $\beta_0$and$\beta_1$.

## 16.3ENDOGENOUS VARIABLES AND EXOGENOUS VARIABLES

There are two types of variables in simultaneous equations models: endogenous and pre-determined variables.

(i) *Endogenous Variables* are determined jointly and interdependently within the model. The endogenous variables are determined by the exogenous variables. We use notation *Y* for endogenous variables.

(ii) *Predetermined Variables*:Predetermined variables are of three types: (i) exogenous variables, (ii) lagged exogenous variables, and (iii) lagged endogenous variables. Exogenous variables are determined outside the equation system. In other words, these variables are given from outside. For example, in the production function of wheat, rainfall is an exogenous variable.

Remember that exogenous variables determine endogenous variables, but endogenous variables do not determine exogenous variables.

In a simultaneous equationsmodel, we can include the lagged values of endogenous variables. For example, $C_t$ is current year consumption and is assumed to be an endogenous variable. Consumption in the previous time period ($C_{t-1}$) is a lagged endogenous variable. Similarly, we can include lagged exogenous variables (for example, $X_{t-1}$) as an explanatory variable in an equation of the simultaneous equations system.

**Notations**

To facilitate our discussion, we introduce the following example:Let us consider a SEM with $M$ endogenous and $K$exogenous variables.The general $M$ equations model in $M$ endogenous, or jointly dependent, variables may be written as Equation (16.13) which is a time dependent SEM:

$$Y_{1t} = \beta_{12}Y_{2t} + \beta_{13}Y_{3t} + \ldots + \beta_{1M}Y_{Mt} + \gamma_{11}X_{1t} + \gamma_{12}X_{2t} + \ldots + \gamma_{1K}X_{Kt} + u_{1t}$$

$$Y_{2t} = \beta_{21}Y_{1t} + \beta_{23}Y_{3t} + \ldots + \beta_{2M}Y_{Mt} + \gamma_{21}X_{1t} + \gamma_{22}X_{2t} + \ldots + \gamma_{2K}X_{Kt} + u_{2t}$$

$$Y_{3t} = \beta_{31}Y_{1t} + \beta_{23}Y_{3t} + \ldots + \beta_{3M}Y_{Mt} + \gamma_{31}X_{1t} + \gamma_{32}X_{2t} + \ldots + \gamma_{3K}X_{Kt} + u_{3t}$$

……………………………………..……………………………………..

$$Y_{Mt} = \beta_{M1}Y_{1t} + \beta_{M2}Y_{2t} + \cdots + \beta_{M,M-1}Y_{M-1,t} + \gamma_{M1}X_{1t} + \gamma_{M2}X_{2t} + \cdots + \gamma_{MK}X_{Kt} + u_{Mt}$$

...(16.13)

where $Y_1$, $Y_2$,…,$Y_M$ are $M$ endogenousvariables

$X_1$, $X_2$,…,$X_K$ are$K$ predetermined variables(one of these $X$ variables may take a value of unity to allow for the intercept term in each equation)

$u_1$, $u_2$,…,$u_M$ are $M$ stochastic disturbances

$t$ =1, …, $T$areTotal number of observations

$\beta$'s are coefficients of the endogenous variables

$\gamma$'s are coefficients of the predetermined variables

As equation (16.13) shows, there are three types of variables: First, we have endogenous variables, whose values are determined within the model. Second, we have predetermined variables, whose values are determined outside the model. Third, we have the stochastic error terms. The endogenous variables are regarded as stochastic whereas the predetermined variables are non-stochastic in nature.

In notations, $X_{1t}$ is a current (present-time) exogenous variable, whereas $X_{1(t-1)}$ is a lagged exogenous variable. Note that$Y_{(t-1)}$ is a lagged endogenous variable with a lag of one time period. Since the value of $Y_{1(t-1)}$ is known at the current time $t$, it is regarded as non-stochastic. Hence, it is considered as a predetermined variable[1]. In short, current exogenous, lagged exogenous, and laggedendogenous variables are considered as predetermined, as their values are not determined by the model in the current time period.

At this point a question may be taking shape in your mind, how do we construct econometric models? What are the variables that should be included in an equation? Two issues are very important. First, such questions are answered by economic theory. The variables in an econometric model are included based on certain economic theory. In some cases, which are not established theory yet, the equations depend upon the logic that the researcher puts forth. Thus, behind the inclusion of a variable in an equation, there is some logic. Second, it is up to the concepts of economic theory and the model builder to specify the variables that are endogenous and that are predetermined. Although non-economic variables (variables not determined by any economic theory), such as temperature and rainfall, are clearly exogenous or predetermined, the model building must be exercised with great care while classifying economic variables as endogenous or predetermined. One must be able to defend i) the classification of variables into endogenous and exogenous, and ii) inclusion of a particular variable in an equation.

**Check Your Progress 1**

1) Explain the meaning of each of the following terms:

   - Endogenous Variables.
   - Exogenous Variables

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   ……………………………………………………………………………
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2) Distinguish between pre-determined variable and exogenous variable.

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3) Write the general form of a simultaneous equation model.

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[1]It is assumed implicitly here that the stochastic disturbances, the $u$'s, are serially uncorrelated. If this is not the case, $Y_{t-1}$ will be correlated with the current period disturbance term $u_t$ .Hence, we cannot treat it as pre-determined.